Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/24/2025
PI Name:	Wargo, Matthew Ph.D.		
Project Title:	Molecular Mechanisms in a Hypervirulent Klebsiella Pneumoniae Strain Isolated from ISS that Subdues Host Defences and Promotes Virulence		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	05405	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC020N0001-FLAGSHIP, OMNIBUS1 Human Research Program: Crew Health Appendix A; Omnibus1-Appendix B
Start Date:	12/01/2022	**End Date:**	11/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2024 per V. Lehman/JSC (Ed., 10/10/23).		
Key Personnel Changes/Previous PI:	The planetary protections reorganization at JPL led to loss of the initial subcontract PI (Singh) and the Co-I. The postdoc (Chander) was elevated to subcontract PI at that time. All of this happened Feb 2024.		
COI Name (Institution):	Singh, Nitin Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC23K0368		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: Microbes colonizing the crew habitat surfaces of spacecraft during long-term spaceflight represent a risk to astronaut health, particularly as immune function decreases with flight duration. A number of the bacteria detected during surface surveillance of the International Space Station (ISS) are reported to be potentially pathogenic to humans, and it is critical to understand their inherent potential to cause infection and measure the impact that simulated microgravity conditions have on their virulence. Changes in virulence due to simulated microgravity have been demonstrated for Salmonella and Serratia, and transcriptome changes indicative of increased virulence have also been seen for Pseudomonas. During recent International Space Station (ISS) surveys, a biocide and multidrug-resistant Klebsiella pneumoniae strain was detected, and its genome points to potential hypervirulence. The virulence potential and impact of simulated microgravity conditions on this ISS isolated K. pneumoniae strain are not yet known. We hypothesize that the K. pneumoniae F3-2P(2*) ISS strain to be hypervirulent in the mammalian cell. Testing the microbe under the Earth and simulated microgravity conditions with whole animal infections will give an insight into whether the increased virulence is innate to the microbe or acquired under microgravity conditions. Aims: 1- Determine the transcriptional, physiologic, and behavioral responses of ISS and Earth-derived clinical K. pneumoniae to simulated microgravity conditions. 2- Measure K. pneumoniae virulence from standard gravity and simulated microgravity conditions using cell-culture and mouse models of infection. Methods: K. pneumoniae F3-2P(2*) and a comparator Earth-derived clinical isolate (KPPR1) will be grown in nutrient broth media and media mimicking the host lung in both standard 1G and simulated microgravity conditions. Simulated microgravity will be established using high-aspect ratio rotating wall vessels, a capability present in both of the collaborating labs. Additionally, 3D clinostate Gravitae will be used to test the effect of simulated microgravity. Transcriptomes will be measured by RNA-Seq using next-generation sequencing for quantifying gene expression, and comparisons will be made between 1G and microgravity conditions. Klebsiella virulence will be measured by assessing biofilm and aggregate formation, gene expression, capsule production, drug resistance, toxicity to human airway epithelial cells, and pathogenesis during mouse infection. Deliverables: We will be able to determine if the ISS-derived K. pneumoniae is hypervirulent and whether simulated microgravity conditions impact its virulence. The study will be the first to examine these behaviors in the ISS-isolated Klebsiella strain and will provide a novel understanding of the transcriptional and physiological changes involved in its response to microgravity. Significance: K. pneumoniae is a nosocomial pathogen capable of infecting healthy humans when exposure is high. Given the relatively high abundance of this pathogen on ISS surfaces (up to 4% of reads) and its described resistance to the conventional ISS biocides, a hypervirulent strain of K. pneumoniae is of a particular risk to humans. This proposed research is a critical step to determine risk scenarios for astronauts and the development of countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Klebsiella pneumoniae is a nosocomial pathogen capable of infecting healthy humans when exposure is high and, at lower levels, anyone with decreased immune function. Given the relatively high abundance of this pathogen on ISS surfaces (up to 4% of reads), its described resistance to the conventional ISS biocides, and the overall suppression of astronaut immune function, hypervirulent strain of K. pneumoniae represents a particularly risk for humans during space flight. The research we are currently carrying out is a critical step to determine risk scenarios for astronauts and the development of countermeasures.
Task Progress:	We were able to complete the Klebsiella strain responses to low-shear simulated microgravity, test the virulence of the International Space Station (ISS) isolate compared to two terrestrial clinical strains, and assess the transcriptional responses of human airway cells to each Klebsiella strain grown in simulated microgravity or control conditions. In sum, we learned a few important things from this study: First, the ISS isolate of Klebsiella pneumoniae is not particularly virulent in the models we have tested. This suggests that it does not pose an acute risk to crew health. Second, each of the three K. pneumoniae strains tested responds differently to exposure to low-shear simulated microgravity. This suggests that future work should take strain into account and that more work is needed to determine if there are strain microgravity responses that can be predicted based on the strain type or genome content. Third, human airway cells respond differently to each K. pneumoniae strain depending on whether they were pre-grown in simulated microgravity or control conditions. However, those genes were generally not the same when comparing among K. pneumoniae strains. Thus, the strain of bacterium used is paramount to potentially extrapolating findings to predict risks and bacterial functions in space. This is something appreciated in the terrestrial clinical setting, particularly for chronic infections. We suggest further experimentation comparing strain responses.
Bibliography Type:	Description: (Last Updated: 12/21/2023)